SIMULATING EVIDENCE ACCUMULATION IN HUMAN DECISION-MAKING: A PYTHON IMPLEMENTATION AND EVALUATION OF THE DRIFT DIFFUSION MODEL.

Research Report

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Research Focus: Computational Modeling of Human Decision-Making

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Abstract

The Drift Diffusion Model (DDM) is a mathematical model used to describe how people make decisions by accumulating evidence over time. This project implemented a simplified version of the model in Python to investigate how evidence strength and decision-boundary settings affect choice accuracy and decision time in a simulated decision-making process. Nine parameter combinations were examined using three drift rates (0.2, 0.5, and 0.8) and three boundary settings (0.5, 1.0, and 1.5). A total of 9,000 trials were simulated, with 1,000 trials for each combination.

The results showed that the proportion of trials classified as correct increased with drift rate across all boundary settings. The highest recorded accuracy was 76.5%, at a drift rate of 0.8 and a boundary setting of 1.5, while the lowest was 50.8%, at a drift rate of 0.2 and a boundary setting of 0.5. Mean decision time increased with the boundary setting, ranging from 0.0955 to 0.6349 seconds across the tested conditions. No trials remained incomplete at the maximum simulation time.

Overall, the results showed a pattern consistent with a speed–accuracy trade-off: higher boundary settings were associated with greater accuracy and longer decision times. However, the model used a simplified scoring rule in which reaching the upper boundary was classified as correct, and changing the boundary setting also changed the starting position. The findings therefore describe the behaviour of this particular simulation rather than establishing how human participants would perform in real decision-making tasks.

Keywords: Drift Diffusion Model, decision-making, evidence accumulation, Python simulation, choice accuracy, decision time.

1. Introduction

1.1 Background of the Study

Human decision-making often involves choosing between alternatives based on available evidence. The strength of the evidence and the amount of information required before making a choice can influence both the accuracy of a decision and the time taken to reach it. Understanding these relationships is an important aspect of studying decision-making processes.

The Drift Diffusion Model (DDM) is a mathematical model used to explain two-choice decision-making as a process of accumulating evidence over time. In the model, evidence gradually accumulates until it reaches one of two decision boundaries, each representing a possible choice. The drift rate represents the average direction and strength of evidence accumulation, while the separation between the decision boundaries determines how much evidence is required before a decision is made.

A higher drift rate generally represents stronger evidence favouring one choice, whereas wider decision-boundary separation requires more evidence before a response can occur. These parameters provide a basis for investigating how evidence strength and decision requirements influence choice accuracy and decision time.

Computational simulation provides a practical way to examine these relationships. Python can be used to implement the model, generate simulated decision trials, and analyse the resulting outcomes under different parameter conditions. This approach allows the behaviour of the model to be investigated systematically without collecting data from human participants.

1.2 Aim and Objectives of the Study

The aim of this study is to implement and evaluate a simulated Drift Diffusion Model in Python to investigate how evidence strength and decision-boundary separation affect choice accuracy and decision time.

The specific objectives are to:

1. Implement a Drift Diffusion Model simulation using Python.

2. Examine the relationship between evidence strength and choice accuracy.

3. Investigate how decision-boundary separation affects decision time and choice accuracy.

4. Compare the simulated outcomes across different combinations of drift rates and decision-boundary values.

1.3 Research Question

The study addresses the following research question:

How do evidence strength and decision-boundary separation affect choice accuracy and decision time in a simulated Drift Diffusion Model?

1.4 Scope of the Study

This study focuses on the computational implementation and evaluation of a Drift Diffusion Model using Python. It examines simulated decision outcomes under different evidence-strength and decision-boundary conditions, with particular attention to choice accuracy and decision time.

The investigation is based on simulated trials rather than observations collected from human participants. Consequently, the findings are interpreted in relation to the behaviour of the implemented model under the conditions examined.

2. Theoretical Background

2.1 The Drift Diffusion Model

The Drift Diffusion Model (DDM) is an evidence-accumulation model used to explain decision-making between two alternatives. It represents a decision as a process in which evidence accumulates over time until it reaches a decision boundary. The model provides a framework for examining the relationship between decision accuracy and the time required to make a choice (Myers et al., 2022).

Unlike an approach that considers only the final choice, the DDM describes the decision process leading to that choice. Evidence accumulation is influenced by the quality of the available information and by the amount of evidence required before a response is made. Because evidence accumulation is assumed to be noisy, the process can vary across individual trials, even when the model's parameters remain unchanged.

2.2 Evidence Accumulation and Drift Rate

Evidence accumulation refers to the gradual gathering of information relevant to a decision. In the DDM, the drift rate represents the average direction and strength of evidence accumulation toward one of the decision boundaries.

A higher drift rate generally represents stronger evidence favouring one alternative. Under the model, stronger evidence tends to produce faster decisions and greater accuracy. A lower drift rate represents weaker or less informative evidence, which can make the decision process slower and less accurate.

These relationships make drift rate an important parameter for investigating how evidence strength influences simulated decision outcomes.

2.3 Decision Boundaries and the Speed–Accuracy Trade-Off

The DDM represents two-choice decisions using two boundaries, with each boundary corresponding to one of the possible responses. A decision is reached when accumulated evidence crosses either boundary.

Boundary separation describes the distance between the decision boundaries. Wider separation requires the evidence-accumulation process to travel farther before a response can be made. This generally increases decision time and can improve accuracy because more evidence is required before a decision is reached. Narrower separation allows decisions to be made with less accumulated evidence, generally producing faster responses but potentially reducing accuracy (Myers et al., 2022).

This relationship illustrates the speed–accuracy trade-off: decisions can be made more quickly by requiring less evidence, while requiring more evidence can improve accuracy at the cost of additional decision time.

2.4 Relevance to the Present Study

The concepts of drift rate and decision-boundary separation provide the theoretical basis for this study. By varying these parameters in a Python simulation, the study investigates how evidence strength and boundary separation relate to choice accuracy and decision time.

The study focuses on simulated outcomes under the tested conditions. Its purpose is to evaluate the behaviour of the implemented model rather than to draw conclusions directly from observations of human participants.

3. Methodology

3.1 Research Design

This project used a computational simulation to investigate how evidence strength and decision-boundary settings affect choice accuracy and decision time in a simplified Drift Diffusion Model (DDM). The simulation was implemented in Python and did not involve human participants or the collection of data from real people.

Two parameters were varied: the drift rate, which represented evidence strength in the simulation, and the boundary setting, which determined the upper decision threshold. The resulting choices and decision times were recorded and analysed to identify patterns across the different parameter combinations.

3.2 Simulation Parameters

Three drift rates and three boundary settings were selected, producing nine unique parameter combinations. Each combination was tested over 1,000 trials, giving a total of 9,000 simulated trials.

Table 1. Simulation parameters

Parameters	Values / Settings
Drift rates	0.2, 0.5, 0.8
Boundary settings	0.5, 1.0, 1.5
Trials per combination	1,000
Numbers of combination	9
Total simulated trials	9,000
Time step	0.01 seconds
Maximum simulation time per trial	10 seconds
Noise parameter	1.0
Random seed	42
Initial position	Half of the selected boundary setting
Lower boundary	0

The random seed was set to 42 to support reproducibility when the simulation is run with the same code and compatible execution environment.

3.3 Simulation Procedure

Each trial began with the decision variable positioned halfway between the lower boundary, set to zero, and the selected upper boundary. At each time step, the decision variable was updated using a drift component and a random noise component.

The drift component moved the decision variable according to the selected drift rate. The noise component introduced random variation using a Gaussian random value scaled by the square root of the time step and the noise parameter.

The simulation checked the decision variable after each update. If it reached or crossed the upper boundary, the trial ended and the choice was recorded as Option A. If it reached or crossed the lower boundary, the trial ended and the choice was recorded as Option B. If neither boundary was reached within the maximum simulation time of 10 seconds, the trial was classified as incomplete.

This procedure was repeated 1,000 times for each of the nine parameter combinations.

3.4 Outcome Measures

Two main outcome measures were used: choice accuracy and mean decision time.

Choice accuracy was calculated as the percentage of completed trials classified as correct. In this implementation, Option A was counted as correct and Option B as incorrect. Accuracy was calculated using the following expression:

Accuracy (%) = Correct trials ÷ (Correct trials + Incorrect trials) × 100

Incomplete trials were excluded from this calculation. This scoring rule was an operational choice for the simulation and should not be interpreted as independent evidence that Option A was objectively correct in a real decision-making task.

Mean decision time was calculated as the average time taken by completed trials to reach either boundary. Incomplete trials, if any, would have been excluded from this calculation. In the recorded results, no trials were incomplete.

3.5 Data Recording and Analysis

The simulation recorded the results for each parameter combination, including the number of trials, correct choices, incorrect choices, incomplete trials, accuracy percentage, and mean decision time.

The summary results were saved in "ddm_results.csv". Individual trial outcomes were saved in "ddm_trial_results.csv". These files allowed the results to be checked against the simulation output.

Two graphs were generated to visualise the relationships between the parameters and the outcome measures. The first graph, Evidence Strength and Choice Accuracy, showed accuracy across the three drift rates for each boundary setting. The second graph, Decision Boundary Separation and Decision Time, showed mean decision time across the boundary settings for each drift rate.

The graphs were used alongside the numerical results to identify patterns across the nine conditions. The resulting observations were then interpreted in relation to the research question, while taking into account the simplifying assumptions of the implementation.

3.6 Reproducibility

The simulation used a fixed random seed of 42. This setting was intended to make the random sequence reproducible when the same code and compatible execution conditions are used. The experiment was defined by fixed parameter values, a consistent trial procedure, and a specified number of trials per condition.

4. Results

4.1 Overview of Simulation Results

The simulation examined how evidence strength (drift rate) and decision-boundary settings affected choice accuracy and decision time. Nine combinations of drift rate and boundary settings were tested, with 1,000 trials conducted for each combination, giving a total of 9,000 simulated trials.

No incomplete trials were recorded. All simulated trials reached either the upper or lower decision boundary within the maximum simulation time of 10 seconds. The results are presented using a summary table and two graphs showing the relationships between the model parameters and the measured outcomes.

4.2 Summary of Numerical Results

Table 2. Summary of simulation results

drift_rate	boundary	trials	correct	incorrect	incomplete	accuracy_percent	mean_decision_time_seconds
0.2	0.5	1000	508	492	0	50.8	0.0965
0.2	1.0	1000	562	438	0	56.2	0.3093
0.2	1.5	1000	593	407	0	59.3	0.6349
0.5	0.5	1000	579	421	0	57.9	0.1012
0.5	1.0	1000	636	364	0	63.6	0.3028
0.5	1.5	1000	688	312	0	68.8	0.6121
0.8	0.5	1000	618	382	0	61.8	0.0955
0.8	1.0	1000	710	290	0	71.0	0.2918
0.8	1.5	1000	765	235	0	76.5	0.5688

Note: In this simulation, Option A was operationally counted as correct and Option B as incorrect. Accuracy was calculated from completed trials, and mean decision time was calculated using completed trials. These labels describe the simulation's scoring procedure and do not establish real-world correctness.

4.3 Evidence Strength and Choice Accuracy

The results showed that choice accuracy increased as the drift rate increased across all three boundary settings. At a boundary setting of 0.5, accuracy rose from 50.8% at a drift rate of

0.2 to 61.8% at a drift rate of 0.8. At a boundary setting of 1.0, accuracy increased from 56.2% to 71.0%. At a boundary setting of 1.5, accuracy increased from 59.3% to 76.5%.

The results also showed that the boundary setting of 1.5 produced the highest accuracy at each drift rate, while the boundary setting of 0.5 produced the lowest. The highest recorded accuracy was 76.5%, obtained at a drift rate of 0.8 and a boundary setting of 1.5. The lowest was 50.8%, obtained at a drift rate of 0.2 and a boundary setting of 0.5.

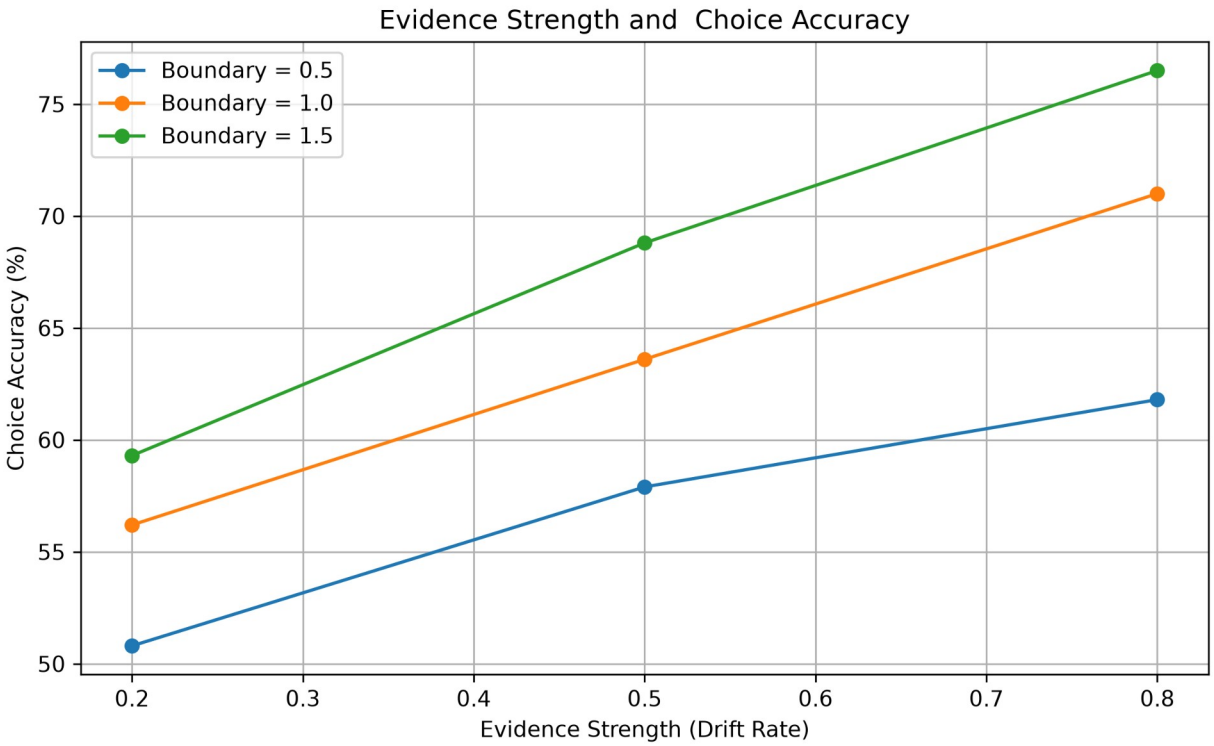


Figure 1. Decision accuracy across different drift rates

Figure 1 illustrates the increase in choice accuracy as evidence strength, represented by the drift rate, increases. All three lines slope upwards, showing this pattern across the tested boundary settings. The line for a boundary setting of 1.5 remains above the other two lines, while the line for 0.5 remains below them. These patterns agree with the numerical results presented in Table 1.

4.4 Decision Boundary Settings and Decision Time

Mean decision time increased as the boundary setting increased across all three drift rates. At a drift rate of 0.2, mean decision time increased from 0.0965 seconds at a boundary setting of 0.5 to 0.6349 seconds at a boundary setting of 1.5. At a drift rate of 0.5, it increased from 0.1012 to 0.6121 seconds. At a drift rate of 0.8, it increased from 0.0955 to 0.5688 seconds.

At the boundary setting of 1.5, the drift rate of 0.2 produced the longest mean decision time, while the drift rate of 0.8 produced the shortest. At the boundary setting of 0.5, however, the mean decision times were all close to 0.1 seconds.

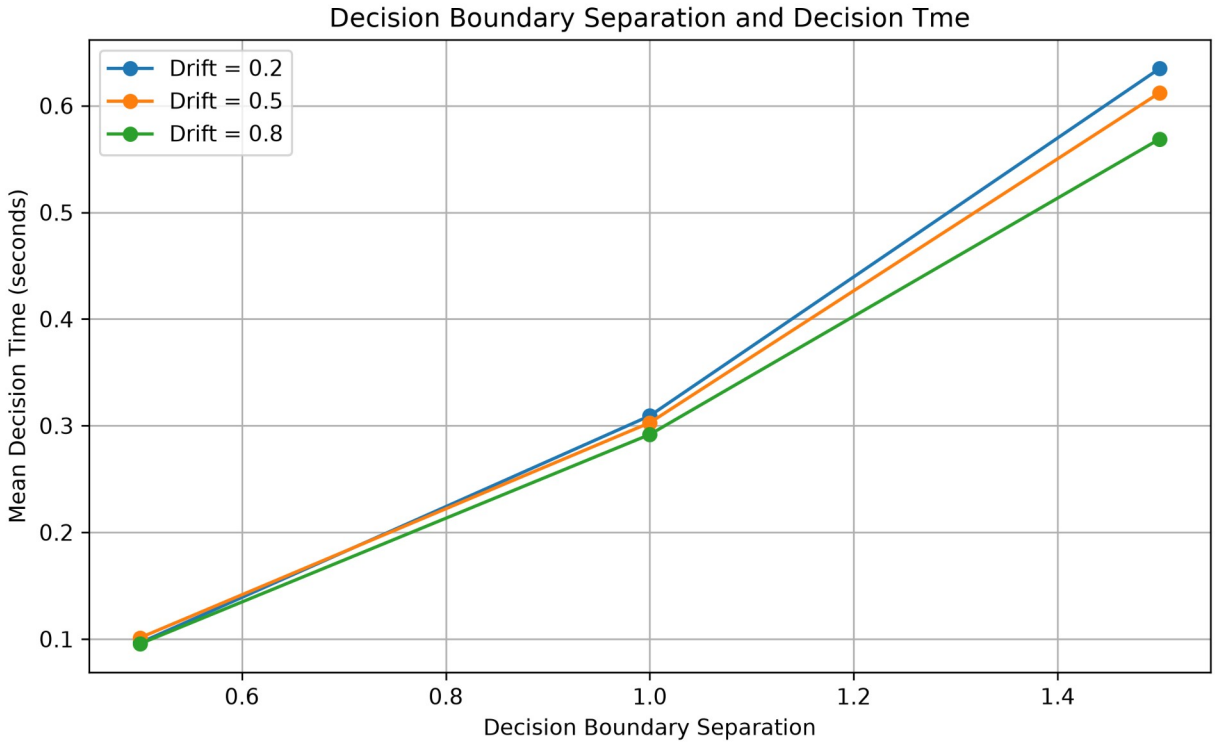


Figure 2. Decision time across different boundary separation

Figure 2 shows the increase in mean decision time as the boundary setting increases. The three lines rise from left to right, indicating that the larger boundary settings were associated with longer decision times in this simulation. The difference between drift rates is more

noticeable at the boundary setting of 1.5 than at 0.5. These visual patterns are consistent with the values reported in Table 1.

4.5 Summary of Key Findings

Three main findings emerged from the simulation. First, choice accuracy increased with drift rate across all tested boundary settings. Second, higher boundary settings were associated with higher choice accuracy. Third, mean decision time increased as the boundary setting increased.

Taken together, these results show a pattern consistent with a speed–accuracy trade-off: the larger boundary settings were associated with greater accuracy but also longer decision times. However, the findings must be interpreted in relation to how the simulation was implemented. In particular, changing the boundary setting also changed the starting position because the starting position was defined as half of the boundary value. Therefore, the results describe the combined effects of these settings in this implementation rather than isolating the effect of boundary separation alone.

5. Discussion

5.1 Overview of the Main Findings

This project investigated how evidence strength and decision-boundary settings affected choice accuracy and decision time in a simplified Drift Diffusion Model implemented in Python. The results revealed three main patterns: accuracy increased with drift rate, higher boundary settings were associated with greater accuracy, and mean decision time increased with the boundary setting.

These patterns were visible in both the numerical results and the two graphs presented in Section 4. Together, they provide an answer to the research question within the limits of the implemented simulation.

5.2 Evidence Strength and Choice Accuracy

The simulation showed that accuracy increased as the drift rate increased across all three boundary settings. For example, at a boundary setting of 1.0, accuracy increased from 56.2%

at a drift rate of 0.2 to 71.0% at a drift rate of 0.8. At a boundary setting of 1.5, accuracy increased from 59.3% to 76.5% across the same drift-rate range.

Figure 1 illustrates this pattern, with all three lines rising as the drift rate increases. Within the simulation, this is consistent with the interpretation that stronger drift moves the decision variable more consistently towards the upper boundary, increasing the proportion of trials classified as correct under the scoring rule used.

However, because Option A was defined as correct and Option B as incorrect, the reported accuracy represents the proportion of simulated choices that reached the designated upper boundary. It does not independently establish the correctness of choices in a real-world task.

5.3 Boundary Settings and Choice Accuracy

The results also showed that higher boundary settings were associated with higher recorded accuracy at every tested drift rate. For example, at a drift rate of 0.5, accuracy increased from 57.9% at a boundary setting of 0.5 to 68.8% at a setting of 1.5.

Figure 1 shows the same pattern: the line representing the boundary setting of 1.5 lies above the lines for the other two settings. This indicates that the proportion of trials reaching the upper boundary was higher under the larger boundary settings in this implementation.

An important qualification is that the boundary setting also determined the initial position of the decision variable. Since the initial position was set to half the selected boundary, changing the boundary setting changed both the upper threshold and the starting position. Consequently, the observed differences cannot be attributed to boundary separation alone.

5.4 Boundary Settings and Decision Time

Mean decision time increased as the boundary setting increased. At a drift rate of 0.2, mean decision time rose from 0.0965 seconds at a boundary setting of 0.5 to 0.6349 seconds at a setting of 1.5. A similar increase occurred at drift rates of 0.5 and 0.8.

Figure 2 visualises this relationship. All three lines rise as the boundary setting increases, indicating that trials generally took longer to reach a decision under the larger settings. At the

boundary setting of 1.5, the mean decision time was 0.6349 seconds for drift 0.2, 0.6121 seconds for drift 0.5, and 0.5688 seconds for drift 0.8.

These results show that, within the simulation, the larger boundary settings were associated with longer decision times. Because the initial position also changed with the boundary setting, however, the observed relationship reflects the combined effect of these implementation choices.

5.5 The Speed–Accuracy Pattern

When the accuracy and decision-time results are considered together, the simulation displays a pattern consistent with a speed–accuracy trade-off. Higher boundary settings were associated with higher recorded accuracy, but they were also associated with longer mean decision times.

This pattern was observed across the three drift rates. For example, at a drift rate of 0.8, accuracy increased from 61.8% to 76.5% as the boundary setting increased from 0.5 to 1.5, while mean decision time increased from 0.0955 to 0.5688 seconds.

The result demonstrates how changing parameters in a computational model can produce a relationship between the time taken to reach a decision and the proportion of simulated choices classified as correct. Nevertheless, the simplified implementation means that this pattern should be interpreted as a property of the present simulation rather than direct evidence about human decision-making.

5.6 Answer to the Research Question

The research question asked: How do evidence strength and decision-boundary separation affect choice accuracy and decision time in a simulated Drift Diffusion Model?

Within this implementation, increasing the drift rate was associated with higher choice accuracy across all tested boundary settings. Increasing the boundary setting was associated with both higher recorded accuracy and longer mean decision time.

These findings answer the research question for the nine parameter combinations examined. However, because the starting position was linked to the boundary setting, the simulation did not isolate the effect of boundary separation independently. Future versions could hold the starting position constant while changing the boundaries to investigate this effect more directly.

6. Limitations

Several limitations should be considered when interpreting the findings of this project.

6.1 Simplified Model Implementation

The simulation used a simplified implementation of the Drift Diffusion Model. Although the decision variable accumulated drift and random noise until it reached one of two boundaries, the implementation did not include every feature or parameter that may be considered in more comprehensive DDM analyses. The findings therefore describe the behaviour of this particular implementation.

6.2 Operational Definition of Accuracy

Option A was counted as correct, while Option B was counted as incorrect. This rule made it possible to calculate a consistent accuracy percentage across the simulated trials. However, the simulation did not use an external source of ground truth to establish that Option A was objectively correct. The reported accuracy should therefore be understood as the proportion of completed trials reaching the designated upper boundary, according to the scoring rule adopted for this project.

6.3 Boundary Setting and Initial Position

The initial position of the decision variable was defined as half of the selected boundary setting. Consequently, increasing the boundary setting changed both the location of the upper boundary and the initial position, while the lower boundary remained fixed at zero.

This means that the observed relationships between boundary settings, accuracy, and decision time cannot be attributed exclusively to changes in boundary separation. The effects of changing the starting position and the upper boundary were not examined independently.

6.4 Absence of Human Participant Data

The project relied entirely on simulated data and did not involve human participants or observations of real decisions. Although the results displayed patterns consistent with evidence accumulation and a speed–accuracy trade-off, they do not establish that real people would show the same numerical results or relationships under actual decision-making conditions.

6.5 Limited Parameter Range

Only three drift rates and three boundary settings were tested, resulting in nine parameter combinations. This allowed the project to examine the selected conditions systematically, but it did not cover the full range of possible parameter values. Other settings could produce different patterns.

6.6 Decision-Time Summary

Mean decision time was calculated using completed trials only. No incomplete trials occurred in the recorded results, so all 9,000 simulated trials contributed to either Option A or Option B outcomes. Nevertheless, the analysis focused on mean decision time and did not provide a more detailed examination of the distribution of decision times.

6.7 Future Improvements

Future work could improve the simulation by varying the starting position independently of the boundary settings, testing a wider range of parameter values, and examining additional features of the decision-time distribution. The implementation could also be compared with suitable empirical data to investigate whether its patterns resemble observed human decision-making.

These improvements would help distinguish the effects of individual parameters and provide a more thorough evaluation of the model's behaviour.

7. Conclusion and Future Work

7.1 Conclusion

This project implemented a simplified Drift Diffusion Model in Python to investigate how evidence strength and decision-boundary settings affected choice accuracy and decision time in a simulated decision-making process. Nine parameter combinations were examined, with 1,000 trials conducted for each combination, resulting in 9,000 simulated trials.

The results showed that recorded choice accuracy increased as the drift rate increased across all three boundary settings. The highest accuracy was 76.5%, recorded at a drift rate of 0.8 and a boundary setting of 1.5. Mean decision time also increased with the boundary setting, reaching 0.6349 seconds at a drift rate of 0.2 and a boundary setting of 1.5. No trials remained incomplete at the maximum simulation time.

When considered together, these findings revealed a pattern consistent with a speed–accuracy trade-off: larger boundary settings were associated with greater recorded accuracy and longer mean decision times. The numerical results and graphs provided a consistent description of these patterns across the tested conditions.

However, the conclusions must be understood in relation to the simulation's design. Option A was operationally classified as correct, and the starting position changed whenever the boundary setting changed. The results therefore describe the behaviour of this specific implementation and do not independently establish the correctness of choices or the behaviour of human participants.

Overall, the project demonstrates how Python can be used to implement a simplified computational model, simulate repeated trials, record outcomes, analyse numerical results, and visualise relationships between model parameters.

7.2 Future Work

Several extensions could improve the simulation and provide a more detailed investigation of evidence accumulation.

Firstly, future work could vary the starting position independently of the boundary settings. This would make it possible to investigate the effect of boundary changes without simultaneously changing the initial position.

Secondly, a wider range of drift rates and boundary settings could be tested to examine whether the observed patterns remain consistent across additional parameter combinations.

Thirdly, future analysis could examine the distribution of decision times rather than relying only on their means. This could provide a more detailed understanding of variation in the time taken to reach a decision.

Finally, the simulation could be compared with suitable empirical data from an established decision-making task. Such a comparison would help evaluate how closely the implementation reproduces patterns observed in actual human behaviour.

These extensions would build on the current project by allowing more controlled parameter comparisons and a more comprehensive evaluation of the model.

References

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